

KAZI ARMAN AHMED

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EDUCATION

Oklahoma State University

Stillwater, OK

Master of Science in Business Analytics and Data Science (BAnDS), **GPA: 4.0**

Aug 2025 - May 2027

Master of Science in Industrial Engineering and Management (IEM), **GPA: 4.0**

Aug 2024 - July 2026

Graduate Certificate in Business Analytics and Data Science (BAnDS), **GPA: 4.0**

Aug 2024 - Dec 2025

- **Honors:** Tau Beta Pi (Engineering Honor Society), Alpha Pi Mu (Industrial Engineering Honor Society)
- **Research:** Stochastic Optimization; Decentralized Energy Systems; Renewable Energy Uncertainty; Machine Learning; Data Analytics; Operations Research.

Military Institute of Science and Technology

Dhaka, Bangladesh

Bachelor of Science in Industrial and Production Engineering (IPE), **GPA: 3.58**

Feb 2016 – Jan 2020

- **Thesis:** Optimization of selective laser sintering three-dimensional printing of thermoplastic polyurethane elastomer: a statistical approach.

WORK EXPERIENCE

Intern, Data Research and Analytics Department (May 2026 – Present)

AAA The Auto Club Group

- Contributed to cloud-based data governance and analytics initiatives by developing Dataplex data quality checks to improve reliability of organizational datasets used in business intelligence and modeling workflows.
- Designed Python and BigQuery-based data extraction workflows to integrate customer data with external economic, weather, fuel price, unemployment, and catastrophe indicators for market attribution analysis.
- Supported development of a market attribution model to evaluate how external factors influence market performance, customer behavior, and strategic decision-making..

Graduate Research Assistant, IEM Department (January 2025 –Present)

Oklahoma State University

- Researching decentralized optimization of renewable-integrated chemical plants using advanced algorithms on High-Performance Computing (HPC) clusters, achieving near-optimal performance and 46% CO₂ reduction.
- Identified critical comorbidity patterns to improve early diagnosis and healthcare outcomes for thousands of patients with Juvenile Idiopathic Arthritis (JIA).
- Developed data preprocessing pipelines for large-scale healthcare datasets and implemented trajectory modeling strategies.

Graduate Teaching Assistant, IEM Department (August 2024 – December 2024)

Oklahoma State University

- Led lab sessions and office hours for ENGR 1412, teaching Python and Excel for data analysis.
- Managed Canvas LMS for 60+ students, including course content updates, exam grading, and discussion forums.
- Received ~95% positive feedback in student evaluations for effective instruction and student engagement.

Data Analyst, Data Management Department (January 2021 – July 2024)

LandQuire Management LLC

- Built automated Power BI dashboards integrating Google Analytics, CRM data, and marketing campaign insights using Python and Supermetrics.
- Reduced manual reporting workload by 85% through scheduled ETL pipelines and report generation scripts and received the Certificate of Excellence award for this contribution.
- Led and trained junior analysts, leading to a 50% increase in team productivity, improved data accuracy by 20%.

Lecturer, Department of Industrial & Production Engineering (June 2020 – Dec 2020)

European University, Bangladesh

- Delivered lectures and supervised undergrad projects in operations research, manufacturing, and optimization techniques.
- Mentored 30+ students on final year capstone projects involving real-world industrial problems.
- Developed new course content incorporating simulation tools and data science principles.

- [P] Richard Reed, **Kazi Arman Ahmed**, Saba Ghasemi, Zheyu Jiang, and Paritosh Ramanan. (2026). **Decentralized Operations of Decarbonized Chemical Plants with Renewable-driven Transmission Systems**. In: *arXiv preprint*.
- [C] **Kazi Arman Ahmed**, Hamid Hadidi, Mohammad Fili, Dursun Delen, Guiping Hu, and Lizhi Wang. (2025). **Comorbidity Patterns Before and After Juvenile Idiopathic Arthritis Diagnosis**. In: *AI, Society and Digital Transformation: Proceedings of 2025 INFORMS Conference on Service Science, University of Oxford, UK, July 1–3, 2025, Lecture Notes in Operations Research, Springer, pp. 276–288*.
- [J] **Kazi Arman Ahmed**, Israt Humaira, Ashiqur Rahman Khan, Md Shamim Hasan, Mukitul Islam, Anik Roy, Mehrab Karim, Mezbah Uddin, Ashique Mohammad, and Md Doulotuzzaman Xames. (2025). **Advancing Breast Cancer Prediction: Comparative Analysis of ML Models and Deep Learning-Based Multi-Model Ensembles on Original and Synthetic Datasets**. In: *PLOS ONE*, 20(6), e0326221.
- [J] Md Mahfuzur Rahman, **Kazi Arman Ahmed**, Mehrab Karim, Jakir Hassan, Rakesh Roy, Bayazid Bustami, S. M. Nur Alam, and Hammad Younes. (2023). **Optimization of Selective Laser Sintering Three-Dimensional Printing of Thermoplastic Polyurethane Elastomer: A Statistical Approach**. In: *Journal of Manufacturing and Materials Processing*, 7(4), 144.
- [J] Anik Roy, Mukitul Islam, Mehrab Karim, **Kazi Arman Ahmed**, Ashiqur Rahman Khan, Mezbah Uddin, and Md Doulotuzzaman Xames. (2023). **Comparative Analysis of KNN and SVM in Multicriteria Inventory Classification Using TOPSIS**. In: *International Journal of Information Technology*, 15(7), 3613–3622.
- [J] Israt Humaira, Kazi Arman Ahmed, Sayantee Roy, Zareen Tasnim Safa, F. M. Tanvir Hasan Raian, and Md. Ashrafuzzaman. (2021). **Design and Development of an Advanced Affordable Wearable Safety Device for Women: Freedom Against Fearsome**. In: *Advances in Science, Technology and Engineering Systems Journal*, 6(2), 829–836.
- [C] Puja Saha, Masuda Afrin Tuba, Kazi Arman Ahmed, and Md. Ashrafuzzaman. (2020). **Development of an Inexpensive Proficient Smart Walking Stick for Visually Impaired Peoples**. In: *Advances in Decision Sciences, Image Processing, Security and Computer Vision, ICETE 2019, Learning and Analytics in Intelligent Systems, Vol. 3, Springer, pp. 48–56*.

HONORS & AWARDS

- **National Champion, Data4Good Competition (Johns Hopkins University) | 2026**
Built a hybrid classification pipeline in Azure to help students detect AI-generated misinformation; featured by CEAT [news](#).
- **1st Runner Up, OSU Datathon (Spears School of Business) | 2026**
Analyzed a large-scale aviation dataset from Textron Aviation to provide prescriptive insights; awarded a scholarship prize and featured in Spears Business [news](#).
- **Creativity, Innovation and Entrepreneurship (CIE) Scholar (Oklahoma State University) | 2026**
Awarded a \$2,500 graduate scholarship recognizing excellence, creativity, innovation, and entrepreneurship; selected to develop a commercialization plan for research-based technology through OSU's NSF I-Corps/Lean LaunchPad process.
- **Winner, MS BAnDS Business Case Competition | 2026**
Recognized by the BAnDS department for excellence in data-driven strategy and business problem-solving.
- **Finalist, IISE Energy Analytics Challenge | 2026**
Selected as a Finalist (Top 5) for the Energy Analytics Challenge, organized by the Energy Division of IISE.
- **Winner, AlphaRamos Marketing Hackathon | 2025**
Applied advanced analytics to solve complex marketing challenges in Dominican Republic; featured in Spears [news](#).
- **SAS Advisory Board Scholarship | 2025–2026**
Awarded for academic excellence and demonstrated potential in the field of Business Analytics and Data Science.
- **Symposium Token Prize Winner, IEM Research Symposium | 2025**
Awarded for the most impactful research presentation in the IEM department based on peer and faculty voting.
- **Certificate of Excellence, LandQuire | 2024**
Recognized for leadership in automating manual reporting processes, increasing data accuracy and operational efficiency.

- **MIST Merit Scholarship, Military Institute of Science and Technology | 2017, 2018**
Ranked in the Top 5 of the academic batch for two consecutive years based on GPA.
- **1st Runner Up, Design & Simulation - National Robotic Festival | 2017**
Recognized at the national level for excellence in robotic system modeling and simulation environments.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **GPSGA Representative, Business Analytics Student Association (BASA) | 2025 – Present**
Serve as the primary liaison between the Graduate Professional Student Government Association (GPSGA) and BANDS program, advocating for student resources and departmental interests.
Participated in the community fundraising initiatives, contributing to resource generation while strengthening engagement with the local Stillwater community.
- **Vice President, INFORMS Student Chapter | 2024 – 2025**
Co-lead the OSU chapter of the leading international association for Operations Research and Analytics.
Arranged 3 professional development workshops and organized 1 research presentation competitions to enhance student engagement by 40%.
- **Volunteer, Bangladesh Student Association (BSA) | 2024 – Present**
Supported 20+ international students in housing, transportation, and academic transition, improving early-stage adjustment and community integration.
- **Scholarly Reviewer | 2024 – Present**
Expert Systems with Applications (Elsevier)
Healthcare Analytics (Elsevier)
Invited to evaluate research papers in AI, optimization, and healthcare data science for top-tier peer-reviewed journals.
- **Advisor, MIST IPE Alumni Association | 2024 – Present**
Organized winter community relief drives and delivered hands-on data science workshops, equipping senior students with practical industry knowledge ahead of their professional careers.
- **Campus Ambassador, Bangladesh Youth Leadership Center (BYLC) | 2019 – 2020**
Selected from a competitive applicant pool to represent a nationally recognized leadership development organization.
Organized leadership training workshops, public speaking sessions, and skill-building events for students.
- **President, MIST Debating Society | Military Institute of Science and Technology | 2019 – 2020**
Directed all club operations, organized national-level debating tournaments, and mentored junior members in critical thinking and public speaking.

PROJECTS

AI Chatbot for Interactive YouTube Queries

- Developed an NLP-powered chatbot that allows users to query specific video timestamps and extract context-aware answers.
- Integrated YouTube transcript parsing and embedding-based similarity search to enhance user interactivity.

Auction Car Price Analysis for GM Financial

- Built end-to-end BI solution with data modeling & dashboards to analyze auction trends and key factors affecting car price.

Power Flow Optimization Web App

- Built an optimization web app reducing generation costs by 15% and visualizing grid operations with auto-layout diagrams.

Real-Time Real Estate Data Visualization using Dash

- Developed a live dashboard with dynamic filtering to visualize scraped property data and monitor real-time KPIs.

Supply vs. Demand Analysis (USA Real Estate)

- Scraped and cleaned nationwide property data to quantify supply-demand gaps, providing actionable insights for investors.

KPI Dashboard for Data Management Department (Tableau)

- Designed an interactive Tableau dashboard to monitor departmental performance which can improve decision-making.

SIMIO-Based Manufacturing Simulation

- Optimized workflow scheduling using simulation, reducing inefficiencies by 90% and ensuring 100% role coverage.

Optimized Work Scheduling for Employee Satisfaction

- Designed a MILP model that reduced scheduling conflict hours by 90% while achieving 100% shift coverage.
- Automated the scheduling process with scalable logic for dynamic labor environments.

SKILLS

- **Programming Languages:** Python, R, SQL, JavaScript, C++

- **Analytical Tools:** Excel, MATLAB, Tableau, Power BI, PySpark, SAS, JMP, Gurobi, Simio, Supply Chain Guru
- **Software:** MS Project, MS Office, MS PowerPoint, Canva, Canvas LMS, Qualtrics, Google Forms, SolidWorks

CERTIFICATIONS

- **AI-900: Microsoft Azure AI Fundamentals**
- **SAS SQL Essentials**
- **IBM Data Analyst Professional Certificate:** A professional certificate program offered by **IBM** through **Coursera**.
- **Certified Data Analyst with Python:** A professional course on advanced uses of python and stats offered by **DataCamp**.
- **MITx's Supply Chain Management MicroMasters®:** offered by Massachusetts Institute of Technology through edX.
- **Specialization Course about Python:** A course about advanced uses of Python offered by “University of Michigan”.
- **Certified Supply Chain Analyst (CSCA):** provided by ISCEA (International Supply Chain Education Alliance).

REFERENCE

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